Supporting Information

Physics-motivated multi-output machine learning for nanozyme kinetics:
predictive utility and limits of cross-publication transfer

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This file provides detailed machine-learning validation, conventional-model benchmarking, feature-group ablation, activity/Arrhenius ablation, and publication-aware out-of-fold summaries for the frozen 86-record cohort. Metrics are evaluated in log10 target space as described in the main manuscript.

Table S1. Conventional tree-model benchmark under shuffled five-fold cross-validation.

Model	Target	Mean R2	SD R2	Mean RMSE log10	Mean MAE log10	Within 2x (%)
Random Forest - base	Km (mM)	-0.171	0.263	1.424	1.146	20.915
Random Forest - base	kcat (s-1)	0.526	0.113	2.201	1.631	13.987
Random Forest - base	Vmax (nM s-1)	0.208	0.255	1.460	0.952	29.216
HistGradientBoosting - base	Km (mM)	-0.141	0.163	1.404	1.166	15.163
HistGradientBoosting - base	kcat (s-1)	0.260	0.102	2.751	2.191	13.987
HistGradientBoosting - base	Vmax (nM s-1)	-0.030	0.093	1.660	1.134	18.627
ExtraTrees - base	Km (mM)	-0.182	0.309	1.426	1.163	16.275
ExtraTrees - base	kcat (s-1)	0.408	0.161	2.445	1.951	9.346
ExtraTrees - base	Vmax (nM s-1)	0.085	0.268	1.554	1.004	31.569
Random Forest - physics-motivated	Km (mM)	0.328	0.379	1.082	0.850	23.203
Random Forest - physics-motivated	kcat (s-1)	0.539	0.093	2.173	1.584	19.804
Random Forest - physics-motivated	Vmax (nM s-1)	0.331	0.072	1.354	0.891	26.797
HistGradientBoosting - physics-motivated	Km (mM)	0.381	0.158	1.025	0.840	17.516
HistGradientBoosting - physics-motivated	kcat (s-1)	0.390	0.089	2.501	1.924	12.810
HistGradientBoosting - physics-motivated	Vmax (nM s-1)	0.197	0.127	1.488	1.030	15.163
ExtraTrees - physics-motivated, independent	Km (mM)	0.346	0.363	1.068	0.811	31.307
ExtraTrees - physics-motivated, independent	kcat (s-1)	0.472	0.111	2.323	1.751	20.850
ExtraTrees - physics-motivated, independent	Vmax (nM s-1)	0.313	0.120	1.348	0.842	35.033
ExtraTrees + RegressorChain - physics-motivated	Km (mM)	0.346	0.363	1.068	0.811	31.307
ExtraTrees + RegressorChain - physics-motivated	kcat (s-1)	0.466	0.106	2.336	1.776	22.026
ExtraTrees + RegressorChain - physics-motivated	Vmax (nM s-1)	0.321	0.116	1.353	0.829	42.092

All models use identical outer folds and train-only preprocessing. Physics-motivated models include the engineered representation described in the main manuscript.

Table S2. Conventional tree-model benchmark under publication-grouped pooled OOF validation.

Model	Target	Pooled OOF R2	OOF RMSE log10	OOF MAE log10	Within 2x (%)	Within 5x (%)
Random Forest - base	Km (mM)	-0.199	1.488	1.257	17.442	24.419
Random Forest - base	kcat (s-1)	0.082	3.108	2.304	17.442	26.744
Random Forest - base	Vmax (nM s-1)	-1.118	2.520	1.821	18.605	34.884
HistGradientBoosting - base	Km (mM)	-0.062	1.400	1.190	16.279	24.419
HistGradientBoosting - base	kcat (s-1)	-0.572	4.067	3.563	2.326	11.628
HistGradientBoosting - base	Vmax (nM s-1)	-0.542	2.150	1.556	17.442	38.372
ExtraTrees - base	Km (mM)	-0.299	1.549	1.302	16.279	23.256
ExtraTrees - base	kcat (s-1)	-0.015	3.268	2.503	10.465	24.419
ExtraTrees - base	Vmax (nM s-1)	-1.135	2.530	1.831	13.953	38.372
Random Forest - physics-motivated	Km (mM)	0.204	1.213	1.031	10.465	38.372
Random Forest - physics-motivated	kcat (s-1)	0.031	3.194	2.458	5.814	16.279
Random Forest - physics-motivated	Vmax (nM s-1)	-0.887	2.378	1.764	13.953	27.907
HistGradientBoosting - physics-motivated	Km (mM)	0.294	1.142	0.943	20.930	44.186
HistGradientBoosting - physics-motivated	kcat (s-1)	-0.348	3.766	3.165	9.302	15.116
HistGradientBoosting - physics-motivated	Vmax (nM s-1)	-0.829	2.341	1.752	18.605	30.233
ExtraTrees - physics-motivated, independent	Km (mM)	0.072	1.309	1.092	9.302	36.047
ExtraTrees - physics-motivated, independent	kcat (s-1)	-0.032	3.295	2.498	15.116	27.907
ExtraTrees - physics-motivated, independent	Vmax (nM s-1)	-0.771	2.304	1.657	18.605	39.535
ExtraTrees + RegressorChain - physics-motivated	Km (mM)	0.072	1.309	1.092	9.302	36.047
ExtraTrees + RegressorChain - physics-motivated	kcat (s-1)	-0.014	3.266	2.552	8.140	19.767
ExtraTrees + RegressorChain - physics-motivated	Vmax (nM s-1)	-0.804	2.325	1.650	25.581	38.372

Each record is predicted exactly once by a model trained without any record from that publication group.

Table S3. Feature-group ablation under shuffled five-fold cross-validation.

Feature set	Target	Mean R2	SD R2	Mean RMSE log10	Mean MAE log10	Within 2x (%)
Raw conditions only	Km (mM)	-0.074	0.136	1.363	1.106	18.627
Raw conditions only	kcat (s-1)	0.283	0.074	2.707	2.356	2.353
Raw conditions only	Vmax (nM s-1)	0.128	0.091	1.537	0.987	27.908
Material descriptors only	Km (mM)	-0.192	0.226	1.436	1.211	9.281
Material descriptors only	kcat (s-1)	-0.258	0.242	3.574	3.262	1.111
Material descriptors only	Vmax (nM s-1)	-0.183	0.392	1.751	1.223	20.980
Raw conditions + material	Km (mM)	-0.182	0.309	1.426	1.163	16.275
Raw conditions + material	kcat (s-1)	0.381	0.157	2.503	2.081	4.641
Raw conditions + material	Vmax (nM s-1)	0.065	0.192	1.583	1.026	24.444
pH family only	Km (mM)	-0.102	0.189	1.382	1.110	20.915
pH family only	kcat (s-1)	0.371	0.062	2.535	1.986	11.699
pH family only	Vmax (nM s-1)	0.259	0.152	1.422	0.902	30.196
Temperature family + Arrhenius	Km (mM)	0.218	0.341	1.169	0.952	22.157
Temperature family + Arrhenius	kcat (s-1)	0.227	0.152	2.804	2.230	10.523
Temperature family + Arrhenius	Vmax (nM s-1)	0.113	0.113	1.536	1.012	29.216
Full without engineered pH	Km (mM)	0.277	0.341	1.126	0.913	15.033
Full without engineered pH	kcat (s-1)	0.353	0.152	2.565	2.027	11.634
Full without engineered pH	Vmax (nM s-1)	0.237	0.146	1.434	0.901	33.856
Full without all pH	Km (mM)	0.151	0.319	1.223	0.998	15.033
Full without all pH	kcat (s-1)	0.264	0.172	2.733	2.205	9.216
Full without all pH	Vmax (nM s-1)	0.072	0.191	1.575	1.028	25.686
Full without material descriptors	Km (mM)	0.381	0.355	1.039	0.802	31.503
Full without material descriptors	kcat (s-1)	0.451	0.095	2.370	1.821	18.758
Full without material descriptors	Vmax (nM s-1)	0.340	0.093	1.335	0.834	37.386
Full without engineered temperature	Km (mM)	-0.179	0.275	1.425	1.141	19.869
Full without engineered temperature	kcat (s-1)	0.436	0.113	2.396	1.885	16.275
Full without engineered temperature	Vmax (nM s-1)	0.198	0.253	1.458	0.918	32.810
Full representation	Km (mM)	0.345	0.363	1.069	0.812	31.307
Full representation	kcat (s-1)	0.465	0.106	2.336	1.777	22.026
Full representation	Vmax (nM s-1)	0.321	0.116	1.353	0.829	42.092

The estimator is fixed to ExtraTrees + RegressorChain so that differences isolate the information contributed by feature groups.

Table S4. Feature-group ablation under publication-grouped pooled OOF validation.

Feature set	Target	Pooled OOF R2	OOF RMSE log10	OOF MAE log10	Within 2x (%)	Within 5x (%)
Raw conditions only	Km (mM)	-0.047	1.390	1.181	13.953	22.093
Raw conditions only	kcat (s-1)	-0.105	3.409	2.954	4.651	8.140
Raw conditions only	Vmax (nM s-1)	-0.504	2.123	1.469	23.256	40.698
Material descriptors only	Km (mM)	-0.327	1.566	1.296	11.628	31.395
Material descriptors only	kcat (s-1)	-0.185	3.531	3.243	1.163	1.163
Material descriptors only	Vmax (nM s-1)	-0.909	2.392	1.653	18.605	37.209
Raw conditions + material	Km (mM)	-0.299	1.549	1.302	16.279	23.256
Raw conditions + material	kcat (s-1)	0.020	3.210	2.581	8.140	18.605
Raw conditions + material	Vmax (nM s-1)	-0.919	2.399	1.643	22.093	45.349
pH family only	Km (mM)	-0.052	1.394	1.187	12.791	24.419
pH family only	kcat (s-1)	-0.010	3.259	2.945	2.326	6.977
pH family only	Vmax (nM s-1)	-0.566	2.167	1.531	12.791	39.535
Temperature family + Arrhenius	Km (mM)	0.031	1.338	1.074	22.093	41.860
Temperature family + Arrhenius	kcat (s-1)	-0.031	3.293	2.687	12.791	22.093
Temperature family + Arrhenius	Vmax (nM s-1)	-0.492	2.115	1.538	22.093	36.047
Full without engineered pH	Km (mM)	-0.051	1.394	1.144	17.442	33.721
Full without engineered pH	kcat (s-1)	0.048	3.165	2.440	9.302	24.419
Full without engineered pH	Vmax (nM s-1)	-0.834	2.345	1.675	19.767	38.372
Full without all pH	Km (mM)	-0.161	1.464	1.157	22.093	40.698
Full without all pH	kcat (s-1)	0.009	3.228	2.667	3.488	13.953
Full without all pH	Vmax (nM s-1)	-0.766	2.301	1.637	20.930	38.372
Full without material descriptors	Km (mM)	0.177	1.233	1.041	9.302	30.233
Full without material descriptors	kcat (s-1)	-0.163	3.497	2.802	13.953	22.093
Full without material descriptors	Vmax (nM s-1)	-0.580	2.176	1.585	19.767	34.884
Full without engineered temperature	Km (mM)	-0.227	1.505	1.280	15.116	24.419
Full without engineered temperature	kcat (s-1)	-0.011	3.261	2.624	12.791	13.953
Full without engineered temperature	Vmax (nM s-1)	-0.897	2.385	1.669	25.581	39.535
Full representation	Km (mM)	0.074	1.307	1.091	9.302	36.047
Full representation	kcat (s-1)	-0.014	3.266	2.550	8.140	19.767
Full representation	Vmax (nM s-1)	-0.802	2.324	1.648	25.581	38.372

Grouped OOF validation assesses source-aware transfer across the 23 reconstructed publication groups and is not an independent external cohort.

Table S5. Same-fold paired R2 differences for the principal architectural and activity/Arrhenius ablations.

Comparison	Target	Mean delta R2	SD delta R2
M3_minus_M1_engineered_representation	Km (mM)	0.527	0.389
M3_minus_M1_engineered_representation	kcat (s-1)	0.063	0.075
M3_minus_M1_engineered_representation	Vmax (nM s-1)	0.228	0.161
M4_minus_M3_chain	Km (mM)	0.000	0.000
M4_minus_M3_chain	kcat (s-1)	-0.006	0.032
M4_minus_M3_chain	Vmax (nM s-1)	0.008	0.058
A1_minus_A0_activity_category	Km (mM)	0.798	0.347
A1_minus_A0_activity_category	kcat (s-1)	-0.054	0.090
A1_minus_A0_activity_category	Vmax (nM s-1)	0.155	0.180
A2_minus_A0_arrhenius	Km (mM)	0.549	0.393
A2_minus_A0_arrhenius	kcat (s-1)	-0.013	0.062
A2_minus_A0_arrhenius	Vmax (nM s-1)	0.113	0.122
A3_minus_A1_arrhenius_beyond_activity	Km (mM)	0.019	0.023
A3_minus_A1_arrhenius_beyond_activity	kcat (s-1)	-0.015	0.009
A3_minus_A1_arrhenius_beyond_activity	Vmax (nM s-1)	-0.013	0.021

Positive values mean that the enhanced model outperformed the corresponding base model on the same outer fold.

Table S6. Activity/Arrhenius sensitivity after excluding unresolved activity annotations (n=77).

Model	Target	Mean R2	SD R2	Mean RMSE log10	Mean MAE log10
A0 physics no arrhenius	Km (mM)	-0.262	0.437	1.460	1.153
A0 physics no arrhenius	kcat (s-1)	0.366	0.117	2.405	1.810
A0 physics no arrhenius	Vmax (nM s-1)	0.257	0.172	1.149	0.695
A1 physics plus activity category	Km (mM)	0.537	0.329	0.831	0.625
A1 physics plus activity category	kcat (s-1)	0.253	0.226	2.591	1.942
A1 physics plus activity category	Vmax (nM s-1)	0.222	0.185	1.159	0.697
A2 physics plus arrhenius	Km (mM)	0.164	0.287	1.194	0.909
A2 physics plus arrhenius	kcat (s-1)	0.312	0.209	2.490	1.905
A2 physics plus arrhenius	Vmax (nM s-1)	0.255	0.158	1.151	0.691
A3 physics plus activity and arrhenius	Km (mM)	0.551	0.310	0.822	0.621
A3 physics plus activity and arrhenius	kcat (s-1)	0.232	0.248	2.625	1.964
A3 physics plus activity and arrhenius	Vmax (nM s-1)	0.195	0.182	1.180	0.717

The qualitative activity-versus-Arrhenius conclusion is retained after excluding the nine records with unresolved activity class.

Table S7. Publication-grouped pooled OOF summary for the M1-M4 and A0-A3 analyses.

Model	Target	Pooled OOF R2	OOF RMSE log10	OOF MAE log10	Within 2x (%)
M1 raw independent	Km (mM)	-0.299	1.549	1.302	16.279
M1 raw independent	kcat (s-1)	-0.015	3.268	2.503	10.465
M1 raw independent	Vmax (nM s-1)	-1.135	2.530	1.831	13.953
M2 raw chain	Km (mM)	-0.299	1.549	1.302	16.279
M2 raw chain	kcat (s-1)	0.020	3.210	2.581	8.140
M2 raw chain	Vmax (nM s-1)	-0.919	2.399	1.643	22.093
M3 physics independent	Km (mM)	0.074	1.308	1.092	9.302
M3 physics independent	kcat (s-1)	-0.032	3.296	2.498	15.116
M3 physics independent	Vmax (nM s-1)	-0.771	2.304	1.656	18.605
M4 full physics chain	Km (mM)	0.074	1.308	1.092	9.302
M4 full physics chain	kcat (s-1)	-0.014	3.266	2.550	9.302
M4 full physics chain	Vmax (nM s-1)	-0.803	2.325	1.650	25.581
A0 physics no arrhenius	Km (mM)	-0.198	1.488	1.265	18.605
A0 physics no arrhenius	kcat (s-1)	-0.076	3.365	2.567	16.279
A0 physics no arrhenius	Vmax (nM s-1)	-0.845	2.352	1.663	26.744
A1 physics plus activity category	Km (mM)	0.168	1.240	1.017	15.116
A1 physics plus activity category	kcat (s-1)	0.001	3.242	2.609	3.488
A1 physics plus activity category	Vmax (nM s-1)	-0.666	2.235	1.571	23.256
A2 physics plus arrhenius	Km (mM)	0.074	1.308	1.092	9.302
A2 physics plus arrhenius	kcat (s-1)	-0.014	3.266	2.550	9.302
A2 physics plus arrhenius	Vmax (nM s-1)	-0.803	2.325	1.650	25.581
A3 physics plus activity and arrhenius	Km (mM)	0.138	1.262	1.021	15.116
A3 physics plus activity and arrhenius	kcat (s-1)	-0.013	3.265	2.605	3.488
A3 physics plus activity and arrhenius	Vmax (nM s-1)	-0.664	2.234	1.572	23.256

These source-aware results complement the fold-wise summaries and show the overall OOF behavior across all 86 records.